Data and Information Quality
Project ID: 13
Dataset: Strutture

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A.Y. 2025-2026

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1 Setup choices

From data exploration, the distribution of the numerical columns of the dataset resulted to be not gaussian, so this information led to using **IQR** and **boxplot** to later detect outlier values. The **assessment process** helped to highlight which columns were the most effort requiring during the later tasks of data cleaning, e.g. *Piani totali* and *Piano piano* showed about 40% of completeness or there was a lack of consistency between *Camere* and *Posti letto*'s values (around 25-30%).

During the **standardization process**, operations were chosen according to the unique values for each column. The comprehension of those values has been fundamental to individuate the right technique to adopt in order to standardize those values, exploiting their already existent structure or using common practices.

Regex was used during different stages – both in **Assessment** and in **Missing Values Handling** – in order to retrieve information from the *Ubicazione* column and use it to compare or insert data. Also, already existent and consistent data – after due inspections – has been used to handle missing values. Though, some missing data couldn't be retrieved, so **standard values** have been **imputed** because computing estimates wouldn't have been realistically correct since the instances are specific and correlated to real world buildings, it would have lead to insert untruthful data.

Instead, during the **Outlier Detection stage** a different assumption has been made for the **consistency correction** of those columns regarding values about elements per floor, i.e. *Piano piano*, *Camere piano*, and *Posti letto per piano*; and their respective total counter, i.e. *Piani totali*, *Camere*, and *Posti letto*. The assumption here is that **some errors have been occurred during the manual insertion** of data into the table. So, some values have been modified during this process in order to make the dataset **consistent between correlated columns**. This operation could be considered risky for **information loss**, but the consistency had the better since its absence would have lead to information loss anyway.

Since there were no exact duplicates in the dataset, during the **Deduplication process** exact matches have been identified considering *Insegna* and the pair *Descrizione via, Civico* as **keys**. Though, the duplicates found using the paired columns as key were many, **Record Linkage** has been adopted as support, filtering those that had **similar values**. After their identification, despite the different *Insegna* values, those which had the **least information** (missing values after standardization) have been discarded.

2 Pipeline implementation

Below the pipeline is described in details following the structure of the notebook.

2.1 Data Quality Assessment and Data Profiling

During this stage, data has been first explored, i.g. what are called the columns and what data they contain. Then, duplicates are checked from the beginning but, as already said in the previous section, no duplicates have been found in the dataset. Furthermore, some metrics have been calculated:

- **Completeness**, to check how many missing values were in the dataset.
- **Accuracy**, just for *ZD* column since other columns didn't have some correct logic to be assumed.
- **Distinctness**, to check how much the values can vary in each column.
- **Consistency**, to check if address columns contained or not the same information of *Ubicazione* column; also, to check correctness of correlation between counters and lists.

Timeliness has not been considered since there were no temporal data in the dataset to use in its computation.

Correlation and **distribution** have been calculated for **numerical columns**. The correlation didn't provide any useful information for the data preparation pipeline, the distribution was **not gaussian** – as evident in the plots – so that information has been used later to check the **outliers** through **boxplot**. The Profile Report shows better the information analyzed in this stage.

2.2 Data Cleaning

2.2.1 Data Transformation & Standardization

The task of this process is to standardize, normalize and transform the values in order make them reusable, understandable and syntactically correct.

The first operation has been to **rename** all the columns in order to remove spaces, use capitalized letters, correct typos and assign a new proper name. After that, **abbreviations** from *TipoVia* columns were replaced with their corresponding long and capitalized version. Also, *TipoStruttura* has been normalized joining two semantically identical values.

Other values for the *Civico* column have been extracted from *Ubicazione* in order to handle more values, since *Civico* column contained some null values. These values have been split in three new temporary columns, two for the number and one for the letters. In later stages, the columns will be joined again properly to insert in the *Civico* column missing and wrong data.

Then, the proper **format** has been set to the columns, so that some will contain strings (e.g. address columns) and other numeric integer values (e.g. counters columns). The *Category* column has been standardized in order to make it a proper category, using a numerical range.

The list columns – i.g. *CamerePerPiano*, *PostiLettoPerPiano*, *ListaPiani* – have been transformed in order to return a single value or a list of values joined by *;* character, depending on the starting data. In particular, the first two above mentioned columns will contain numeric values; instead, the last one will contain string values since floors can be identified also by letters and the sequence has been ordered according to their semantic value.

2.2.2 Error detection and correction

Handling missing values The task of this stage is to identify and locate the missing values and use proper techniques to handle and replace them.

First, since address columns contains some null values, they have been retrieved from the *Ubicazione* column that contained redundant data. Indeed, after this operation, *Ubicazione* column got dropped from the table.

After that, it's time to **retrieve missing data** about *CodiceVia* from other rows that have the same *NomeVia* value. Luckily, in the dataset there were some **multiple instances** for the same street and they were not inconsistent, allowing this handling process, though some values couldn't be retrieved and they will be **imputated** later. Unfortunately, this couldn't be done with the missing values of *ZD* columns, since there were no multiple instances of streets with the same name.

Then, missing values for *PianiTotali* column has been retrieved from *ListaPiani*, counting the elements from the corresponding list. This couldn't be done for *Camere* and *PostiLetto* since the only rows, that contained missing values for these columns, contained other missing values from the respective list-format columns.

At the end, after all the possible operations to retrieve existent data from the dataset, all the missing values have been imputated with their respective **standard values**.

Outliers correction Here the goal is to identify, evaluate and correct potential outliers.

First, outliers for *Civico* values have been evaluated through **boxplot**, but only one value has been identified as true outlier (it has been individuated from the **assessment stage** already). Since it was just one value, it has been corrected with its true one after it's been discovered making a brief research.

Other **boxplots** showed potential outliers for *Camere*, *PianiTotali* and *PostiLetto* values, but they have been evaluated as not outliers after their manual inspection.

Here it comes the **consistency check** between *ListaPiani*, *CamerePerPiano* and *PostiLettoPerPiano*: since they all represent lists of elements per floor, they should have the same length. The constraint here was to check if *CamerePerPiano* and *PostiLettoPerPiano* lists contained less or equal elements than *ListaPiani* (the assumption is that not all floors are intended for rooms). If the constraint was not respected, then – depending on the situation – the floor list got increased or decreased (coherently with the list structure described above), or the last elements of beds/rooms lists got removed to make the constraint valid.

After that, another consistency check was of course made for the respective counters columns, i.g. *PianiTotali*, *Camere*, *PostiLetto*, to ensure that their values corresponded respectively to the new lengths of the lists. This transformation of data could be meant as information loss or alteration, but during this task more importance has been given to the **consistency of the dataset** in order for it to not contain errors or ambiguous data. As alternative, it could have easily invalidated the value indicating its not availability.

2.2.3 Data Deduplication

The task of this stage is to detect exact duplicates or potential ones based on their similarity.

Since there were no **exact duplicates**, it's been checked the **exact matches** for the *Insegna* values, first, and for the address columns, later.

5 replicas out of 6 – one was just a false positive, same name but different address – have been individuated through exact matching of *Insegna* values and removed from the table, the choice has been based on the **information quantity** of the rows.

Instead, for the subtask using the address values as key research for duplicated, **Record Linkage** has been used as a support to select those row who also had **similar values** in *Insegna*, *Categoria*, *TipoStruttura*, *Camere*, *PostiLetto*. 12 couples have been detected and 12 rows have been dropped, according to the **least informative** ones.

2.3 Data Quality Re-Assessment

After all the data preparation operations, a **second brief quality assessment** has been done in order to guarantee effectiveness of the work done on the dataset.

The final results are indeed excellent, since all the metrics, the same calculated previously during the first assessment task, show a **100%** of **completeness** and **consistency**. A new **Profile Report** is available for better visualization.